



Unit No 3

Understanding the Interaction

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Unit No 3 A Glance



- Understanding device specific interactions and human aspects involved
- **Interaction Styles-** Direct Manipulation and Virtual Environments, Menu Selection, Form Filling and Dialog Boxes, Command and Natural Languages, Interaction Devices, Collaboration and social media Participation
- Interacting with voice, visual and audio-visual interfaces
- Understanding Scenarios and context of use for user.
- Exercise - Listing down scenarios for an application/system, critical task list for each scenario Understanding user journey and user journey maps.
- Exercise - Creating a user journey map for a particular task

Understanding device specific interactions and human aspects involved



Revisiting HCI Definition



What Is Human-Computer Interaction (HCI)?

- HCI is the study of how people interact with computers, especially as it relates to technology design. User-centered design, **UI**, and **UX** are combined with HCI to provide intuitive technology and products.
- HCI specialists consider how to develop and deploy computer systems that satisfy human users. The majority of this research focuses on enhancing human-computer interaction by enhancing how people utilize and comprehend an interface.



Understanding device-specific interactions and the human aspects involved

1. Device-Specific Interactions

- **Interface Design:** Each device has its own interface considerations. For example, mobile phones use touch screens, while desktops use a mouse and keyboard. Designing for each interface requires different approaches to usability and ergonomics.
- **Input Methods:** Devices can have varied input methods such as touch, voice, gestures, or physical buttons. Ensuring that the device supports intuitive and efficient input methods for the user is important.
- **Screen Size and Resolution:** The size and resolution of the screen affect how content is displayed. Design must adapt to various sizes, ensuring readability and usability.
- **Performance and Resources:** Devices have different performance capabilities and resource limitations (e.g., processing power, memory). Optimizing applications to run efficiently on different devices is essential.
- **Connectivity:** Different devices may have different connectivity options (Wi-Fi, Bluetooth, cellular networks). Designing for varying connectivity scenarios can impact user experience.

Understanding device-specific interactions and the human aspects involved

▣ **Touchscreens (Smartphones, Tablets):**

- **Gestures:** Swiping, pinching, tapping, and other gestures that are native to touch devices.
- **Responsive Design:** Interfaces that adapt to different screen sizes and orientations.
- **Input Methods:** Virtual keyboards, voice commands, and stylus inputs.

▣ **Desktops and Laptops:**

- **Keyboard and Mouse:** Traditional point-and-click, drag-and-drop, and keyboard shortcuts.
- **Precision:** Higher precision for detailed tasks like design, coding, and content creation.

Understanding device-specific interactions and the human aspects involved

▣ **Wearables (Smartwatches, Fitness Trackers):**

- **Small Screens:** Limited screen space requires simplified interfaces.
- **Contextual Interactions:** Often rely on contextual awareness, like location or movement.
- **Quick Interactions:** Focus on quick, glanceable information and interactions.

▣ **Voice-Activated Devices (Smart Speakers, Voice Assistants):**

- **Natural Language Processing:** Understanding and responding to voice commands.
- **Hands-Free Operation:** Interactions that are entirely voice-driven, requiring no physical input.

▣ **AR/VR Devices:**

- **Immersive Interactions:** Requires 3D navigation and interaction with virtual objects.
- **Spatial Awareness:** Understanding the user's physical space and movements to enhance immersion.

Understanding device-specific interactions and the human aspects involved

2. Human Aspects Involved

- **Cognitive Load:** Devices should be designed to minimize cognitive load, making tasks easier to understand and complete. This includes clear instructions, intuitive navigation, and reducing unnecessary complexity.
- **Ergonomics:** The physical design of a device should accommodate the human body to reduce strain and discomfort. This includes screen placement, button sizes, and overall ease of use.
- **Accessibility:** Devices and applications should be accessible to people with disabilities. This includes providing alternative input methods, screen readers, and customizable settings for visual and auditory impairments.
- **Behavioral Patterns:** Understanding how people interact with devices, including their habits and preferences, can guide the design of more user-friendly technology. This includes recognizing common use cases and typical user flows.
- **Emotional Response:** The design should consider the emotional impact of the device on users. This can include aesthetic elements, feedback mechanisms, and the overall user experience.

Understanding device-specific interactions and the human aspects involved

3. Interaction Design Principles

- **Consistency:** Maintain consistent design elements across devices to help users feel familiar with the interface.
- **Feedback:** Provide clear feedback for user actions to help them understand the results of their interactions.
- **Flexibility:** Allow users to customize their experience and adapt the device to their needs and preferences.
- **Error Prevention:** Design to prevent user errors and provide clear guidance for correcting mistakes

Understanding device-specific interactions and the human aspects involved

4. Cross-Device Considerations:

- **Consistency:** Ensuring a consistent user experience across different devices.
- **Seamless Transition:** Allowing users to start a task on one device and continue it on another without interruption.
- **Adaptive Design:** Interfaces that adapt not just to the device, but also to the user's needs and preferences.

5. User Testing and Feedback

- **Usability Testing:** Regularly test the device with real users to identify issues and areas for improvement. This can help ensure that the device meets user needs and expectations.
- **Iterative Design:** Use feedback to iteratively improve the design, making adjustments based on user experiences and evolving needs.

How many computers/ Interaction Devices ...

in your house?

- PC
 - TV, VCR, DVD, HiFi, cable/satellite TV
 - microwave, cooker, washing machine
 - central heating
 - security system
- can you think of more?

in your pockets?

- PDA
 - phone, camera
 - smart card, card with magnetic strip?
 - electronic car key
 - USB memory
- try your pockets and bags

Text entry devices

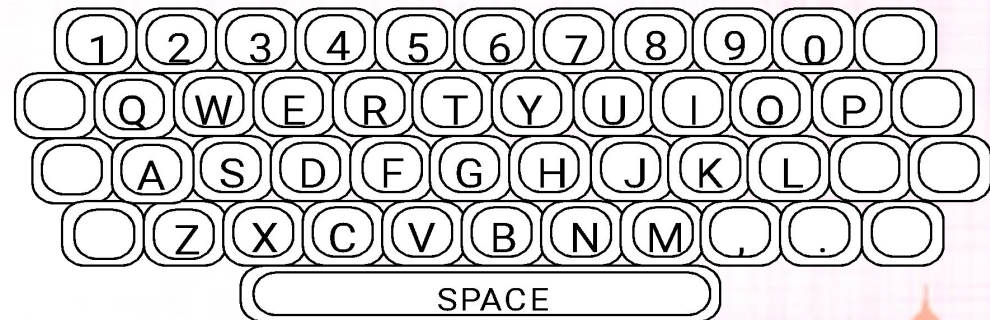
keyboards (QWERTY – Common Keyboards)
chord keyboards, phone pads
handwriting, speech



Keyboards

- Most common text input device
- Allows rapid entry of text by experienced users
- Keypress closes connection, causing a character code to be sent
- Usually connected by cable, but can be wireless

layout – QWERTY



- Standardized layout
but ...
 - non-alphanumeric keys are placed differently
 - minor differences between UK and USA keyboards
- QWERTY arrangement not optimal for typing
 - layout to prevent typewriters jamming!
- Alternative designs allow faster typing but large social base of QWERTY typists produces reluctance to change.



alternative keyboard layouts

Alphabetic

- keys arranged in alphabetic order
- not faster for trained typists
- not faster for beginners either!

Dvorak

- common letters under dominant fingers
- biased towards right hand
- common combinations of letters alternate between hands
- 10-15% improvement in speed and reduction in fatigue



e.g. the Maltron left-handed keyboard

Chord keyboards

only a few keys - four or 5

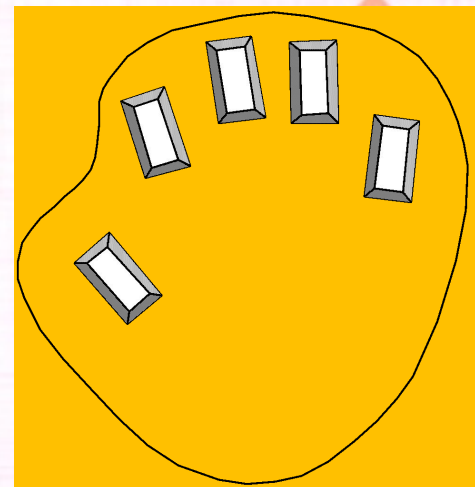
letters typed as combination of keypresses

compact size

ideal for portable applications

short learning time

keypresses reflect letter shape fast once you have trained



phone pad and T9 entry



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- use numeric keys with multiple presses

2 - a b c 6 - m n o
3 - d e f 7 - p q r s
4 - g h i 8 - t u v
5 - j k l 9 - w x y z

hello = 4433555[pause]555666

surprisingly fast!

- T9 predictive entry
 - type as if single key for each letter
 - use dictionary to 'guess' the right word
 - hello = 43556 ...
 - but 26 -> menu 'am' or 'an'



Handwriting recognition

- Text can be input into the computer, using a pen and a digitizing tablet
 - natural interaction
- Technical problems:
 - capturing all useful information - stroke path, pressure, etc. in a natural manner
 - segmenting joined up writing into individual letters
 - interpreting individual letters
 - coping with different styles of handwriting
- Used in PDAs, and tablet computers ...
... leave the keyboard on the desk!



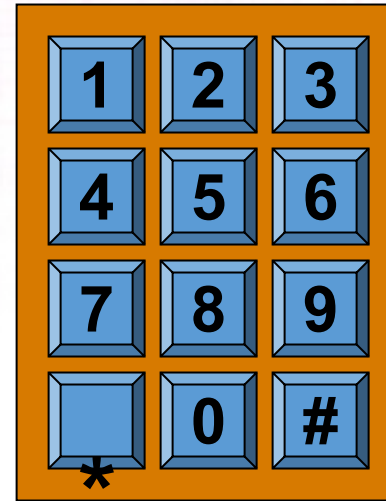
Speech recognition

- Improving rapidly
- Most successful when:
 - single user – initial training and learns peculiarities
 - limited vocabulary systems
- Problems with
 - external noise interfering
 - imprecision of pronunciation
 - large vocabularies
 - different speakers

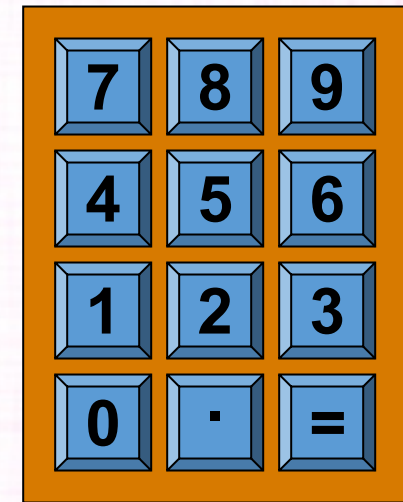


Numeric keypads

- for entering numbers quickly:
 - calculator, PC keyboard
- for telephones



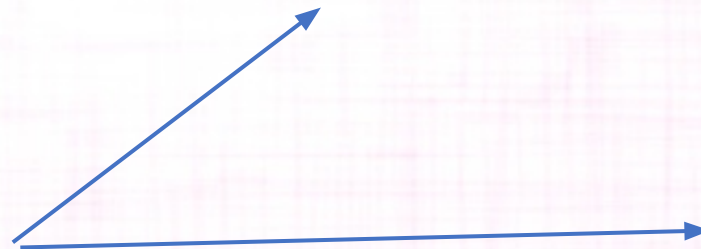
telephone



calculator

Is there Significant Difference ?

Ans= YES, of course



not the same!!

ATM like phone



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Visual and audio-visual interfaces



Audio connectors and video connectors are electrical or optical connectors for carrying audio or video signals. Audio interfaces or video interfaces define physical parameters and interpretation of signals.

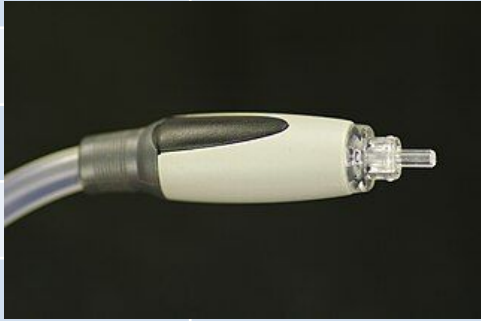
For digital audio and digital video, this can be thought of as defining the physical layer, data link layer, and most or all of the application layer.



Interfaces and their connectors



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Audio or video	Digital or analog	Description	Connectors
Audio only	Analog	Often unmarked on consumer audio equipment since it is so common, or labelled with headphones symbol or as "line out". Computers and other equipment sometimes use Microsoft-Intel color coding scheme, especially when there are multiple input/output plugs.	RCA connector 
	Digital	S/PDIF (Sony/Philips Digital Interconnect Format). Via coaxial or optical cables.	
		 TOSLINK	 BNC  3.5 mm TRS minijack  XLR Connector

Interfaces and their connectors



Audio or video	Digital or analog	Description	Connectors
Video only	Analog	Video Graphics Array (VGA)	D-subminiature 15-pin
		Composite. Often designated by the CVBS acronym, meaning "Color, Video, Blank and Sync".	RCA jack, normally yellow (often accompanied with red and white for right and left audio channels respectively)
		S-Video (Separate Video). Carries standard definition video and does not carry audio on the same cable.	Mini-DIN 4-pin
		Component. In popular use, it refers to a type of analog video information that is transmitted or stored as three separate signals. Either RGB interfaces or YPbPr	3 RCA jacks
		Composite, S-Video, and Component	VIVO = Mini-DIN 9-pin with breakout cable.
	Digital and analog	Digital Visual Interface (DVI)	DVI connector

Interfaces and their connectors

Audio or video	Digital or analog	Description	Connectors
Video and audio	Analog	SCART (Peritel)	SCART
	Digital	High-Definition Multimedia Interface (HDMI), BNC	HDMI connector
		DisplayPort	DisplayPort connector
		HDBaseT	8P8C connector
		IEEE 1394 "FireWire"	FireWire or i.LINK connectors



Interfaces and their connectors



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Interaction styles



Interaction Styles:

- 1. Command line:** The user types in commands for the program, usually one at a time. The program executes the commands and returns feedback, if necessary. MS-DOS and UNIX use this style.
- 2. Question and answer:** The application asks questions and when the user provide by answers all necessary data, the application gives the results.
- 3. Menus:** Possible user actions are listed on the screen and the user can select one of them. Gopher is an example, and most MS Windows applications also include menus.
- 4. Form filing:** The user types the data in specific fields, like the fields on a paper fill-in form. Many office and database applications use this style.
- 5. Function keys:** The interaction is done by a set of special keys or key combinations for different operations. Typical examples are computerized ticket machines, computer games, etc.
- 6. Graphical direct manipulation:** The objects used in application are graphically represented on the screen and the user can manipulate them directly by pointing, clicking, dragging, typing, etc. Most windowing systems, or GUI (Graphical User Interface) are based on graphical direct manipulation.

Direct Manipulation:

- Direct manipulation is an interaction style where users act on displayed objects of interest using physical, incremental, and reversible actions. The effects of these actions are immediately visible on the screen. Key characteristics of direct manipulation interfaces:
 - Continuous representation of objects of interest that users can interact with
 - Physical actions like clicking, dragging, and gesturing instead of complex syntax
 - Incremental, reversible actions with continuous feedback
 - Immediate visibility of results, allowing easy error detection and correction
- Examples of direct manipulation include:
 - Dragging a file icon to move it between folders
 - Pinch-zooming to resize an image on a touchscreen
- Direct manipulation reduces cognitive load by aligning actions with real-world experiences. It promotes learning through exploration and experimentation.

What is “Direct Manipulation”?

Direct manipulation is a human–computer interaction style which involves continuous representation of objects of interest and rapid, reversible, and incremental actions and feedback.

There are so many examples of direct-manipulation systems such as :

- Command line vs. display editors and word processors (The advances of WYSIWYG word processors like Microsoft Word)
- Video games
- Computer-aided design (CAD)
- Office automation

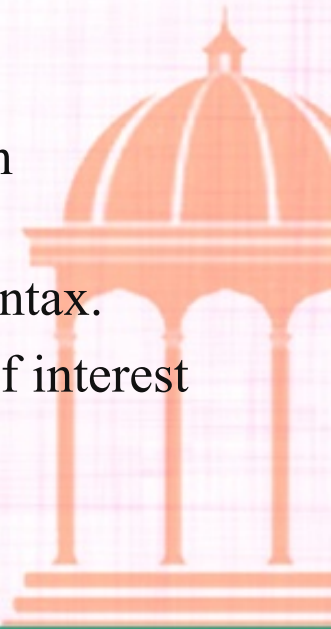


Direct manipulation is closely associated with interfaces that use windows, icons, menus, and a pointing device (WIMP GUI) as these almost always incorporate direct manipulation to at least some degree. However, direct manipulation should not be confused with these other terms, as it does not imply the use of windows or even graphical output.

For example, direct manipulation concepts can be applied to interfaces for blind or vision-impaired users, using a combination of tactile and sonic devices and software.

Principles of direct manipulation :

- Continuous representations of the objects and actions of interest with meaningful visual metaphors.
- Physical actions or presses of labeled buttons, instead of complex syntax.
- Rapid, incremental, reversible actions whose effects on the objects of interest are visible immediately.



What is interaction?



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communication

user ↔ m

Models of interaction

- terms of interaction
- Norman model
- interaction framework

Some terms of interaction

domain – the area of work under study
e.g. graphic design

goal – what you want to achieve
e.g. create a solid red triangle

task – how you go about doing it
– ultimately in terms of operations or actions
e.g. ... select fill tool, click over triangle



Natural Language Interaction (NLI)



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Natural language interaction refers to the ability of computers to understand and respond to human language

- **Example of Natural Language Interaction:**
- **Virtual Assistants**
 - Virtual assistants like Siri, Alexa, and Google Assistant use NLI to understand voice commands and respond in natural language. For example, you can ask Alexa "What's the weather today?" and it will interpret your query, fetch the weather data, and provide a conversational response.
- **Chatbots**
 - Chatbots leverage NLP to engage users in natural language conversations, whether via text or voice. They can understand queries, provide relevant information, and even handle simple tasks like scheduling appointments



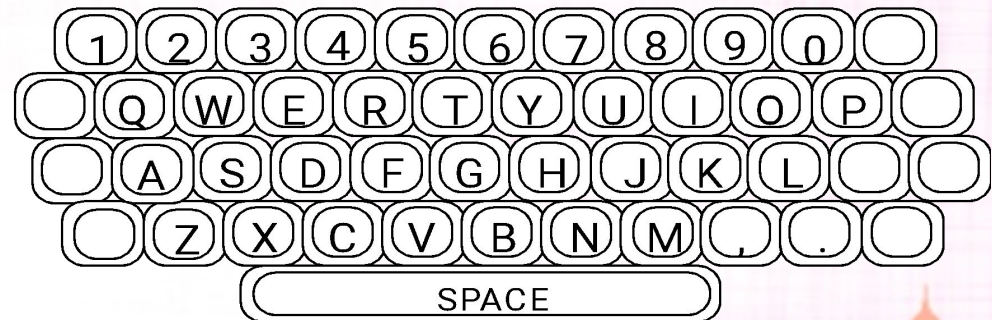
Interaction devices are hardware components that enable users to interact with digital interfaces, programs, or machines. They facilitate input, control, navigation, manipulation, and communication between humans and computers.



Keyboards

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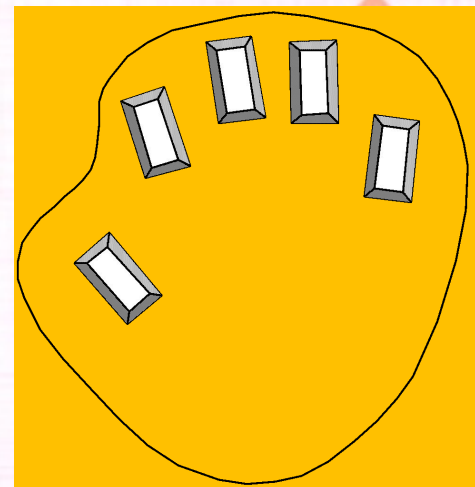
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Pointing Devices

- **Mouse** - Allows cursor control and selection of on-screen elements
- **Touchscreen** - Enables direct interaction by touching the screen surface
- **Trackpad/Touchpad** - Found on laptops for cursor control
- **Trackball** - Allows cursor control by rolling a ball



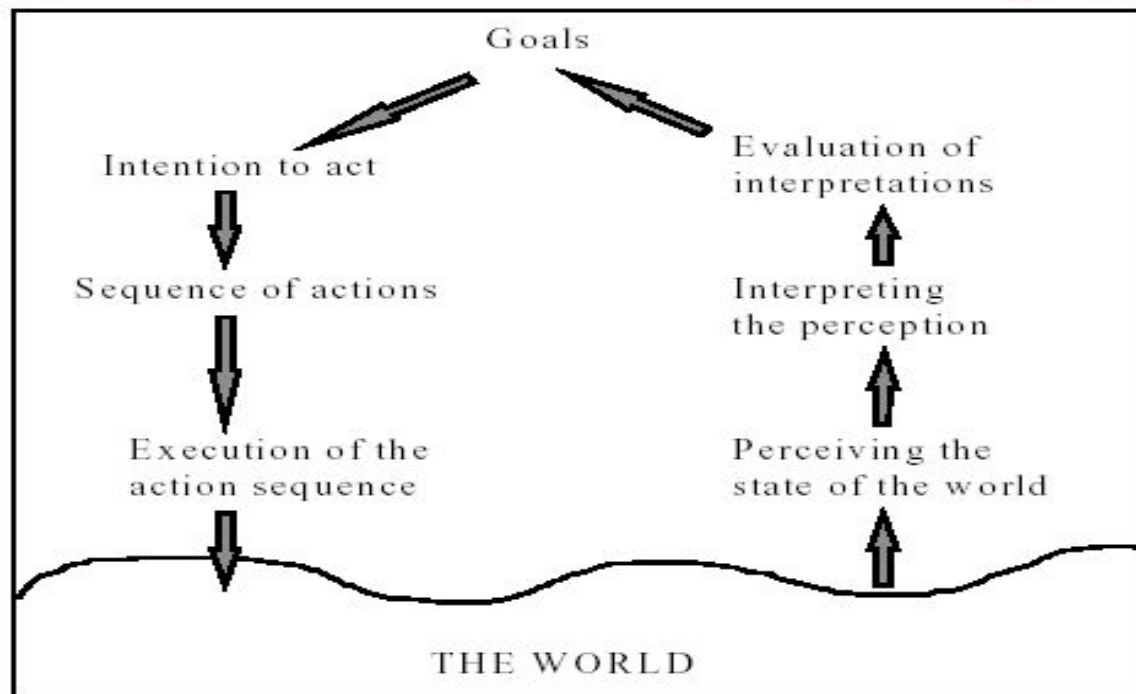
Speech and Auditory Interfaces

- **Speech recognition** - Transforms spoken words into text or commands
- **Voice interfaces** - Enable hands-free interaction using speech recognition and generation
- **Other Devices**
 - Gesture recognition - Converts human gestures and movements into input commands
 - Eye-tracking - Allows control by tracking eye movements
 - Data gloves - Capture hand and finger motions for interaction
 - Haptic devices - Provide force feedback for a more immersive experience
 - Game controllers - Specialized input devices for gaming



Donald Norman's model

- Seven stages
 - user establishes the goal
 - formulates intention
 - specifies actions at interface
 - executes action
 - perceives system state
 - interprets system state
 - evaluates system state with respect to goal



Donald Norman's model

1. **Establishing the goal:** The user identifies what they want to achieve.
2. **Formulating the intention:** The user translates their goal into a specific intention or plan of action.
3. **Specifying the action:** The user determines the actions needed to fulfil their intention.
4. **Executing the action:** The user performs the specified actions.
5. **Perceiving the system state:** The user observes the system's response to their actions.
6. **Interpreting the system state:** The user makes sense of the feedback received from the system.
7. **Evaluating the outcome:** The user assesses whether the outcome meets their original goal and decides on the next steps, which may involve repeating the cycle with a new goal if necessary.

Donald Norman's model: Example

- 1. Establishing the Goal:** The user wants to save their sketch in a specific folder named "ABCD"
- 2. Formulating the Intention:** The user decides they need to select the "Save" option from the file menu to achieve their goal.
- 3. Specifying the Action:** The user identifies the action needed, which is to click on the "Save" label in the menu.
- 4. Executing the Action:** The user performs the action by clicking on "Save."
- 5. Perceiving the System State:** The user observes a progress bar indicating that the file is being saved.
- 6. Interpreting the System State:** The user interprets the completion of the progress bar and notices that the save menu has disappeared, indicating that the action was successful.
- 7. Evaluating the Outcome:** The user checks the specified folder "ABCD" to confirm that the document has been saved successfully.



Collaboration and social media Participation

- Collaboration and social media participation are increasingly recognized as vital components of modern interaction, both personally and professionally.
- The integration of social media into collaborative efforts has transformed how individuals and organizations communicate, share information, and work together.





Goals of Collaboration and Participation

The primary goals of collaboration through social media can be categorized as follows:

- **Focused Partnerships:** Collaborating with specific individuals or organizations to achieve mutual objectives.
- **Structured Work Processes:** Utilizing social media tools to streamline workflows and enhance productivity.
- **Community Engagement:** Building and maintaining online communities that foster interaction and support among members.
- **Information Sharing:** Leveraging social media platforms to disseminate knowledge and resources effectively.





Characteristics of Social Media Participation

Social media participation varies widely, with users engaging in different ways, from passive consumption to active contribution. Key characteristics include:

- **Asynchronous and Synchronous Interactions:** social media allows for both real-time conversations (e.g., video calls) and delayed exchanges (e.g., emails, forum posts), accommodating diverse communication preferences and schedules.
- **Lightweight vs. Substantial Engagement:** Many users engage in "lightweight" interactions, such as liking or sharing content, while others develop deeper relationships through consistent contributions and collaboration.
- **Community Dynamics:** Online communities often consist of lurkers (those who consume content without contributing) and active participants who drive discussions and initiatives.



Benefits of Social Media Collaboration

Engaging in social media collaboration offers numerous advantages:

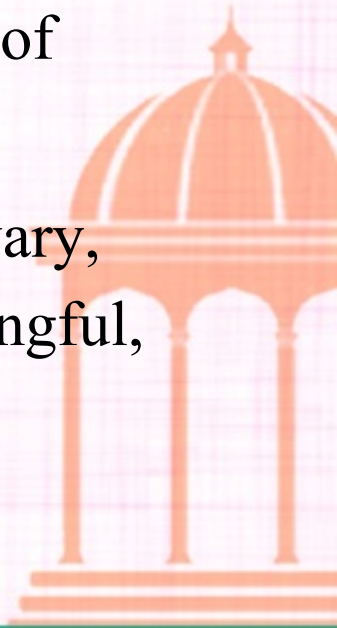
- **Expanded Reach:** Collaborating with others allows individuals and brands to tap into new audiences, increasing visibility and engagement.
- **Enhanced Credibility:** Partnerships with reputable brands or influencers can bolster trust and authority in each field.
- **Cost-Effective Marketing:** Sharing resources for marketing efforts can reduce costs and enhance the effectiveness of campaigns.
- **Creative Synergy:** Collaborations often lead to innovative ideas and content creation, benefiting from diverse perspectives and skills.



Challenges and Considerations

While collaboration through social media presents many opportunities, it also poses challenges:

- **Overload and Distraction:** The sheer volume of information and interactions can lead to cognitive overload, making it difficult for users to focus on essential tasks.
- **Ownership and Control Issues:** Collaborative efforts can raise questions about intellectual property and the management of shared content.
- **Quality of Interactions:** The quality of engagement can vary, with some interactions being superficial rather than meaningful, potentially undermining the collaborative spirit



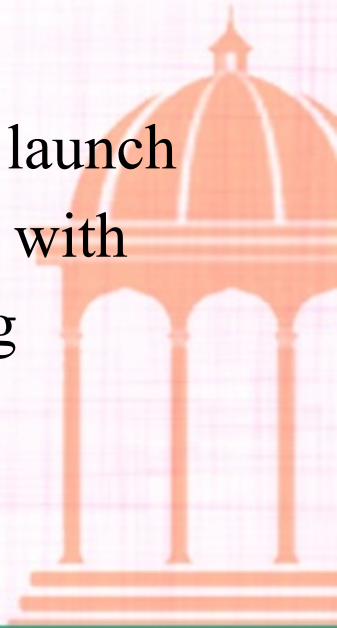
Example:

Brand Collaborations

- **Starbucks and Spotify:** This partnership allowed Starbucks customers to influence in-store playlists through Spotify. This collaboration not only enhanced the customer experience but also creatively merged the two brands' audiences, generating excitement and engagement within their customer bases

Cross-Industry Collaborations

- **Apple and Mastercard:** These two giants collaborated to launch the Apple Card, combining Apple's technological prowess with Mastercard's financial expertise to create a groundbreaking product that appealed to a wide audience



Understanding Scenarios and context of use for user



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- User scenarios are a powerful tool for understanding user behavior and designing effective products or services.
- They help teams visualize a user's journey and steps to achieve a specific goal or resolve a problem.
- By focusing on the "why" rather than just the "how", user scenarios provide valuable insights into users' motivations, intent, and context.



Key Elements of User Scenarios

- An effective user scenario should include the following elements:
- **User Persona:** Describe the user's demographics, personal traits, and relevant habits.
- **Motivator:** Identify the problem or reason why the user needs a specific solution.
- **Intent:** Define the goal the user wants to achieve.
- **Action:** Outline the steps the user will follow to reach their goal.
- **Resolution:** Explain how the user reaches a solution and why they choose it.



Types of User Scenarios

There are three main types of user scenarios:

1. Task-based Scenarios: Focus solely on the user's steps to complete a specific action without additional context.
2. Elaborated Scenarios: Include an elaboration of each step taken by the user, an analysis of their Behavior, and suggestions for improving the user experience.
3. Full-scale Task Scenarios: Combine task-based and elaborated scenarios while including steps for improving the overall user experience



Importance of Context in User Scenarios

Understanding the user's context is crucial for designing effective scenarios. Context includes factors such as:

- **Location:** Where the user is when interacting with the product or service.
- **Device:** The type of device the user is using (e.g., desktop, mobile, tablet).
- **Time:** The time of day or week when the user is engaging with the product.
- **Emotional State:** The user's mood or emotional state during the interaction.
- **Social Setting:** The social environment surrounding the user's interaction.

Benefits of User Scenarios

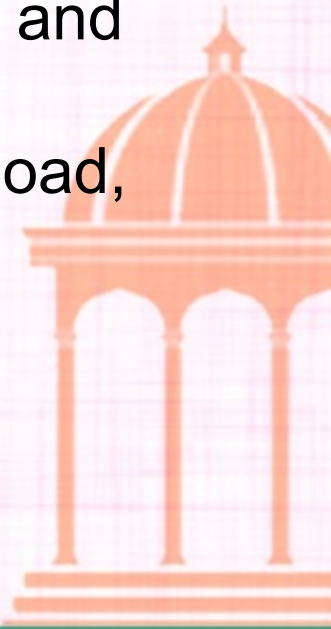
1. **Improved User Understanding:** User scenarios help teams empathize with users and gain a deeper understanding of their needs and motivations.
2. **Alignment and Communication:** User scenarios provide a shared reference point for teams to align on user goals and communicate effectively.
3. **Informed Design Decisions:** User scenarios inform design decisions by highlighting key user needs and pain points.
4. **Iterative Improvement:** User scenarios can be used throughout the design process to continuously refine and improve the user experience.



Example of Scenarios

Elderly User with a Health Monitoring App

- **Goal:** Monitor health metrics like blood pressure, heart rate, and medication reminders.
- **Scenario:** An elderly person uses a health monitoring app on a tablet. The interface must be simple, with large buttons and voice-assisted features. The user can set medication reminders and share health data with their doctor effortlessly. The design prioritizes ease of use, minimal navigation, and clear feedback.
- **Key Considerations:** Accessibility, minimal cognitive load, easy navigation, large fonts, voice integration.



Example of Scenarios

Student Using an Online Learning Platform

- **Goal:** Access course materials, complete assignments, and communicate with instructors.
- **Scenario:** A student logs into an online learning platform, needing to find study materials quickly, submit assignments, and join live discussions. The interface should be structured logically, with easy navigation to various course sections. The system also needs to provide timely feedback on progress.
- **Key Considerations:** Clear navigation, feedback mechanisms, multimedia support, mobile-friendly design.

•





- **Exercise & Case Study- Creating a user journey map for a particular task**

You are asked to Manage Tour of Friends team of 20 to 25 in quantity. Write a scenario for hunting various place for trek through social media applications and plan of tour by registration of team through online mod.

- Listing down scenarios for an application/system, critical task list for each scenario Understanding user journey and user journey maps.



THANK YOU

